

Literature Status for arXiv:1309.3501 Problems 1 and 2

FA/Banach automatic math run

2026-07-03

Status

Status: literature already answered, for the intended graph formulation of the intrinsic-metric and algebra-characterization questions.

Source Question

The source paper is Agelos Georgakopoulos, Sebastian Haeseler, Matthias Keller, Daniel Lenz, and Radoslaw K. Wojciechowski, *Graphs of finite measure*, arXiv:1309.3501 [1]. In Section 8.3, page 53, the paper asks:

Problem 1. Does (D) imply (C) in general?

Here (D) is the condition that all finite-energy functions are bounded, also called canonical compactifiability, and (C) is total boundedness with respect to every intrinsic metric associated with a finite measure. The same page also asks:

Problem 2. Does the fact that $\tilde{D}$ is an algebra imply that (b, c) is canonically compactifiable?

Supporting Answer

The supporting paper is Simon Puchert, *Characterizations of canonically compactifiable graphs via intrinsic metrics and algebraic properties*, arXiv:1803.01942 [2]. Its abstract states that it answers questions from [1].

For Problem 1, Puchert's Theorem 1 proves that if a graph (X, b) is canonically compactifiable and σ is intrinsic with respect to a finite measure, then (X, σ) is totally bounded. Together with the converse from [1], Corollary 1 states that a graph is canonically compactifiable if and only if it is totally bounded with respect to every finite-measure intrinsic metric.

For Problem 2, Puchert's Theorem 2 proves that if the finite-energy space $D(G)$ is an algebra, then the graph G is canonically compactifiable. Together with the converse already known from [1], Corollary 2 states that G is canonically compactifiable if and only if $D(G)$ is an algebra.

Scope

Puchert formulates the result for graphs $G = (X, b)$, the standard zero-killing setting of the metric characterization. This is the formulation Puchert explicitly identifies as answering the questions from [1]. This note does not claim that every possible arbitrary-killing-term reading of the source notation (b, c) is covered.

Search Bounds

The target was checked against the run registry, solution index, attempt index, and proof-gap index for arXiv:1309.3501 and core phrases including “canonically compactifiable”, “finite measure intrinsic metric”, and “finite energy algebra”. Web search for the exact source title and these phrases found arXiv:1803.01942 as the explicit later answer. No competing later source was needed for the identification.

Human Review Recommendation

Mark Problems 1 and 2 from arXiv:1309.3501 as already answered in later literature for the intended graph formulation. Future work should only reopen this target if it explicitly studies a variant involving arbitrary killing terms.

References

- [1] A. Georgakopoulos, S. Haeseler, M. Keller, D. Lenz, and R. K. Wojciechowski, *Graphs of finite measure*, arXiv:1309.3501.
- [2] S. Puchert, *Characterizations of canonically compactifiable graphs via intrinsic metrics and algebraic properties*, arXiv:1803.01942.